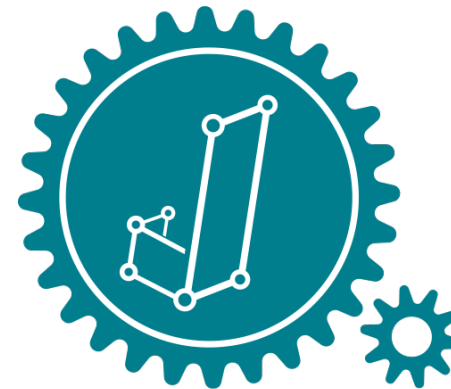
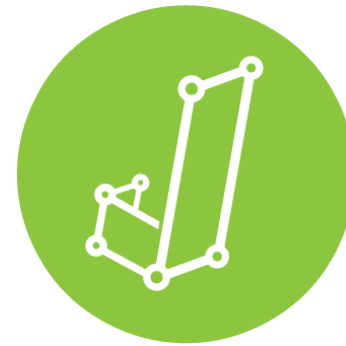


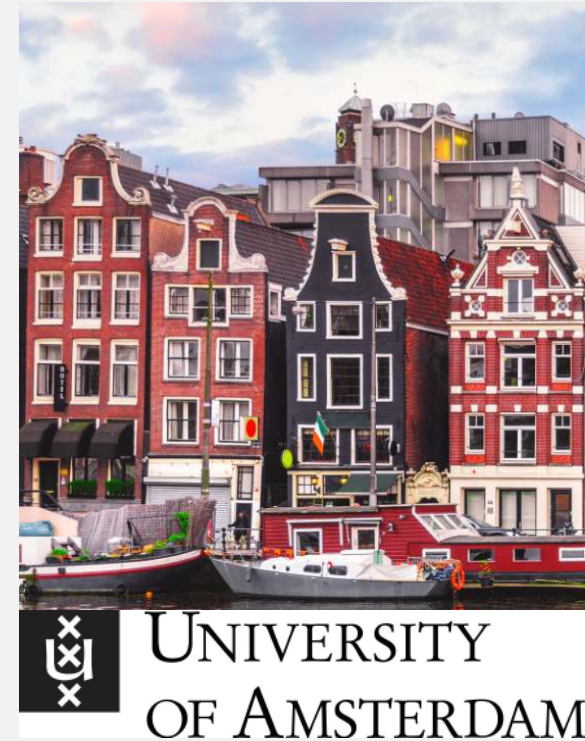
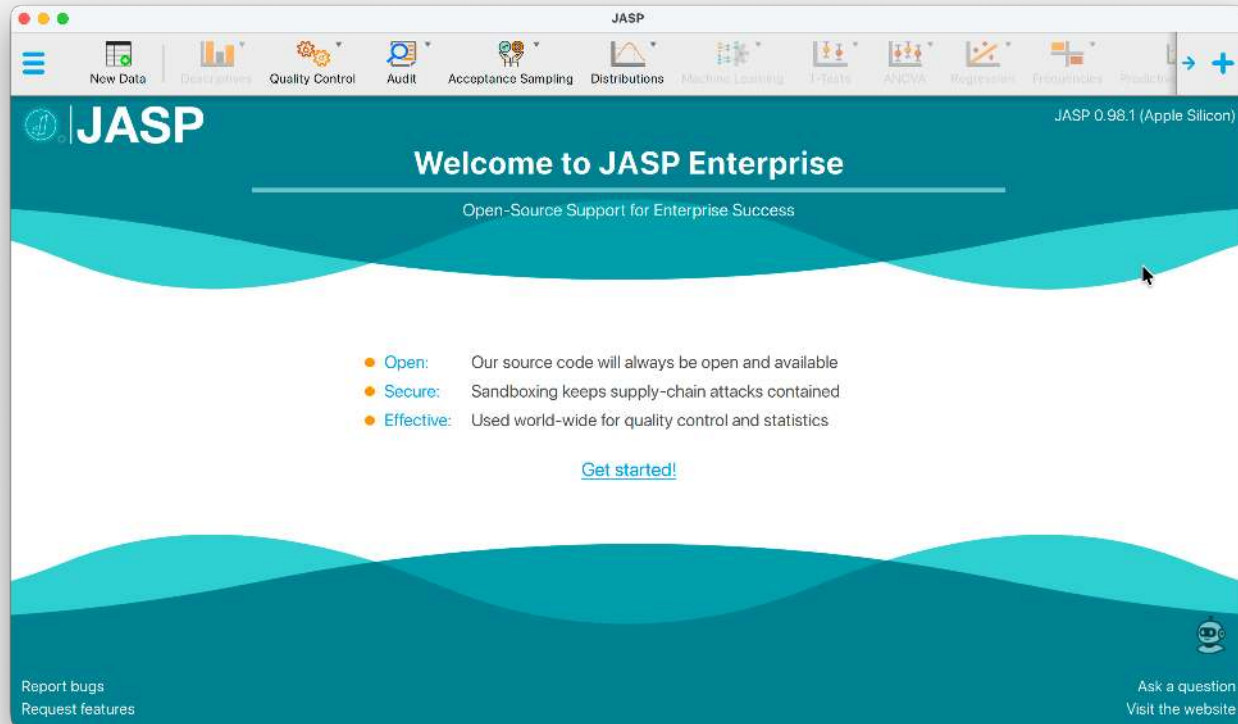
JASP

JASP for Quality Control: An Open-Source
Software for Industrial Statistics

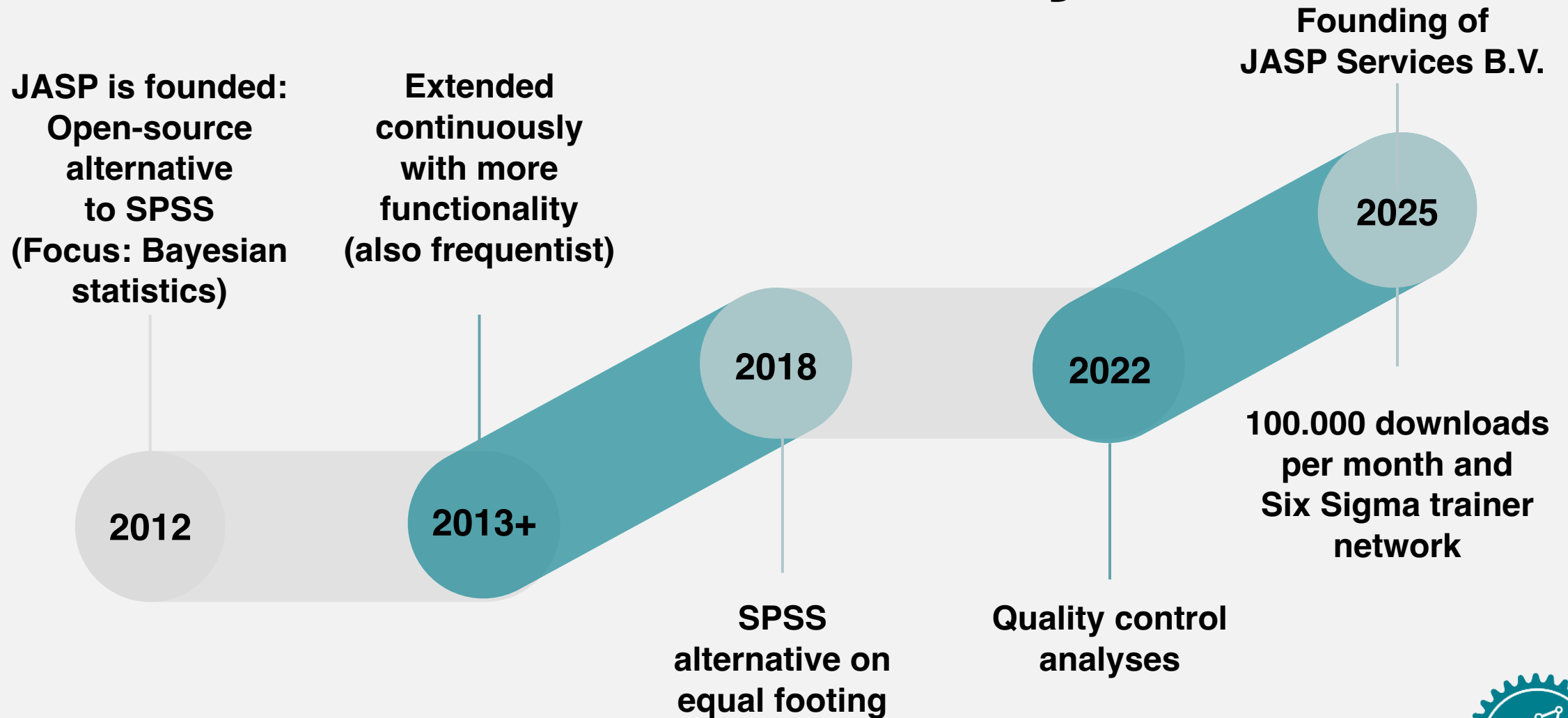
Julius M. Pfadt (PhD) & JASP Team
ENBIS Conference 2026, Florence
Sept 8, 2026



JASP



A Brief History



The Software

AI Integration

Operates the software
Uses your own model
Exposes MCP server hook



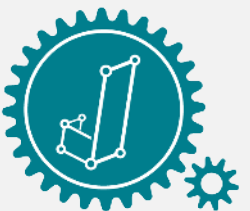
JASP

State-of-the-art
Bayesian and classical analyses
Free: No license fees, no cap on installations and seats



R

Open source statistical foundation



JASP Services B.V.

Support behind the software



**Support &
Contracts**



**Custom
Analyses**



**Batch
Processing**



**Easy
Migration**



**Team
Workshops**



Functionality Today

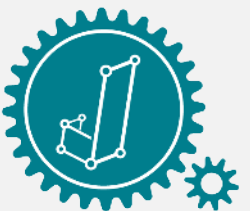
JASP Layout

Data Visualization

Compare Distributions

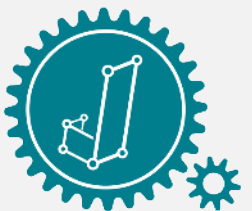
Bayesian Process Capability

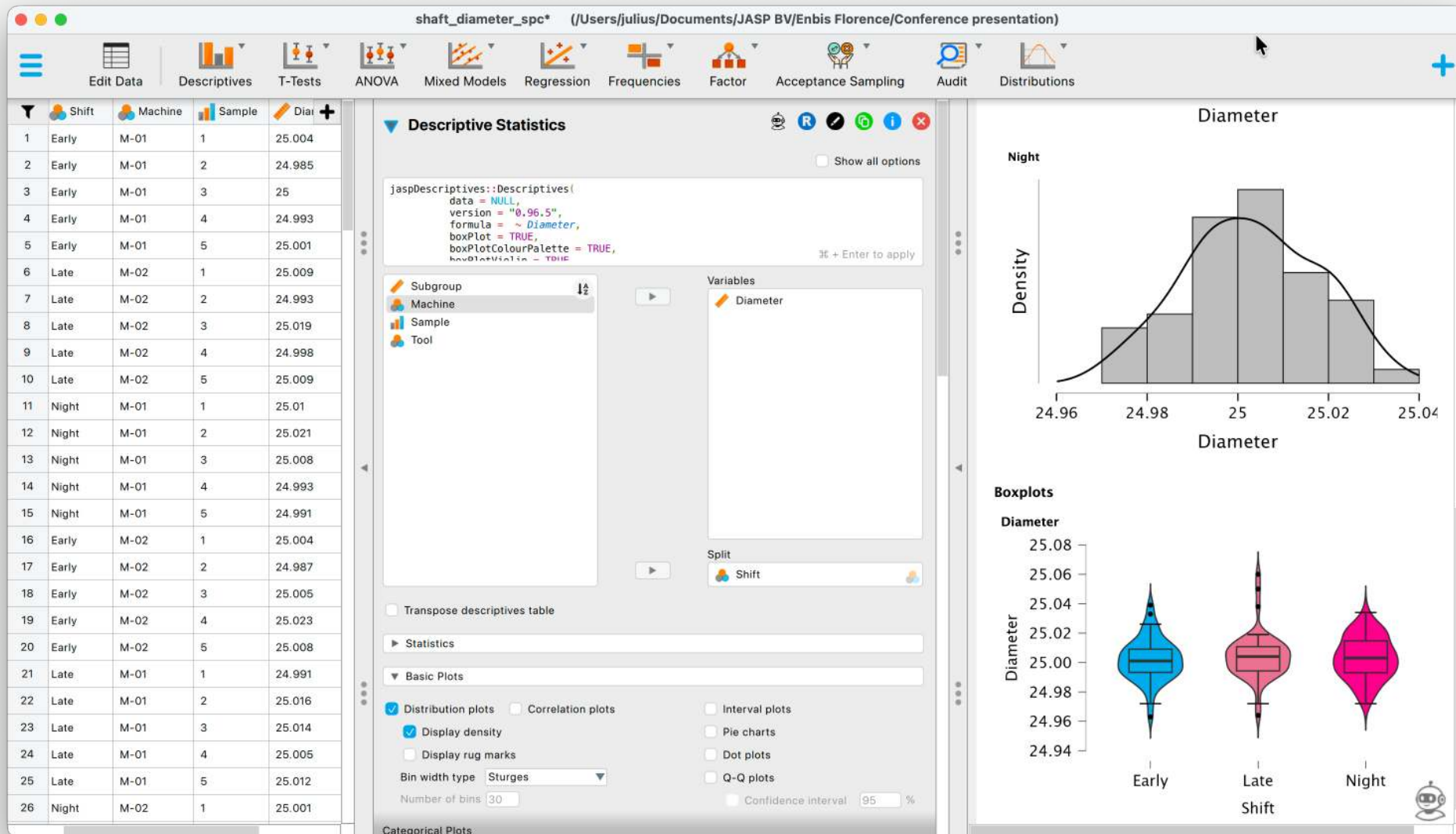
JASP AI





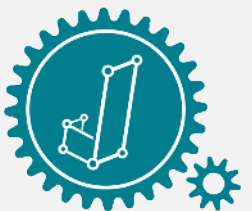
JASP Layout







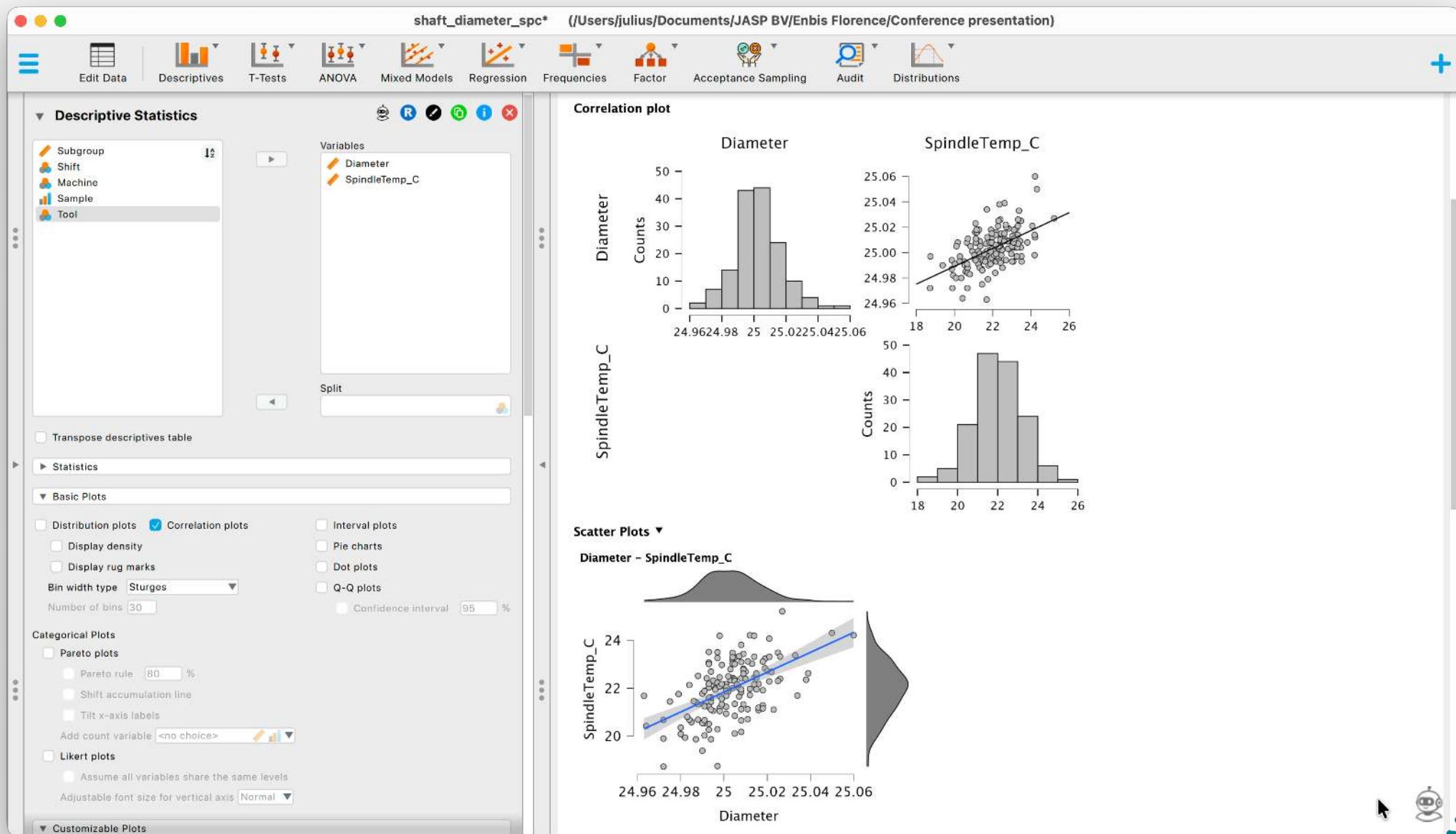
Data Visualization in JASP



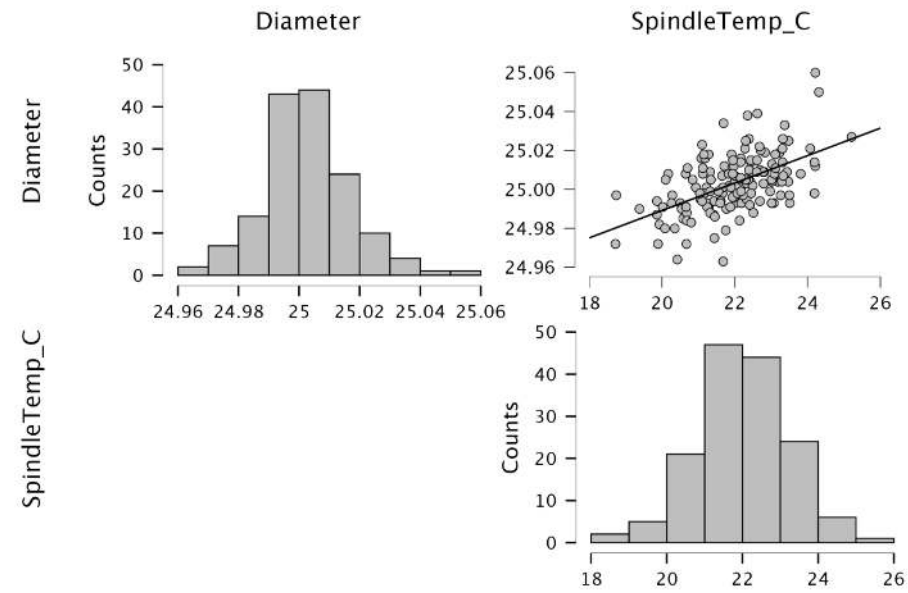
Example: Production of a Shaft

- Target diameter 25mm
- Tolerance $\pm 0.05\text{mm}$
- Secondary variables: Shift, Machine, Roundness, SpindelTemp

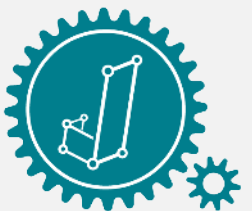
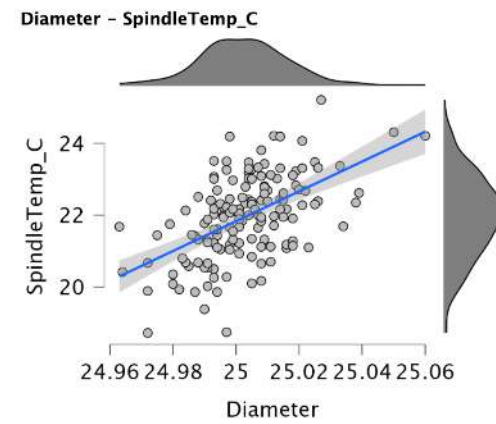


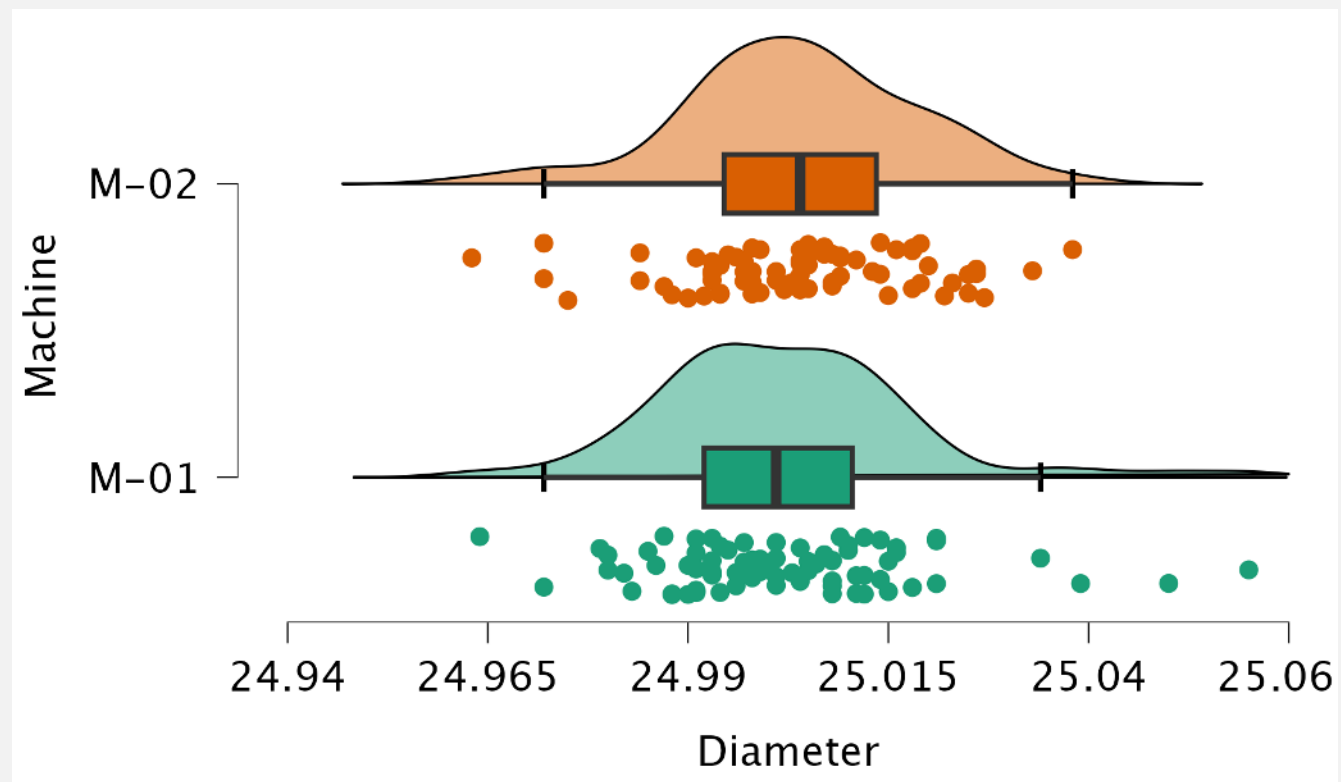


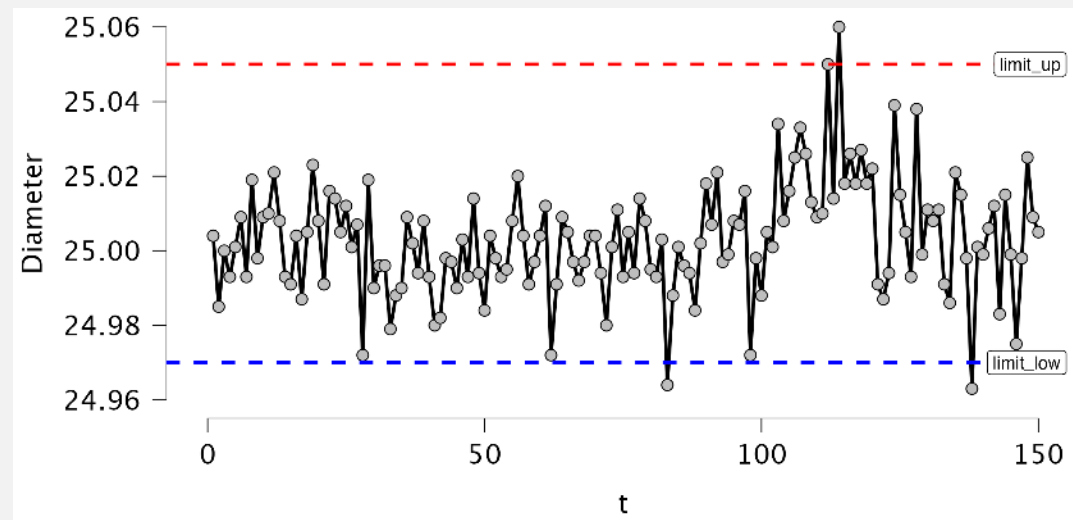
Correlation plot

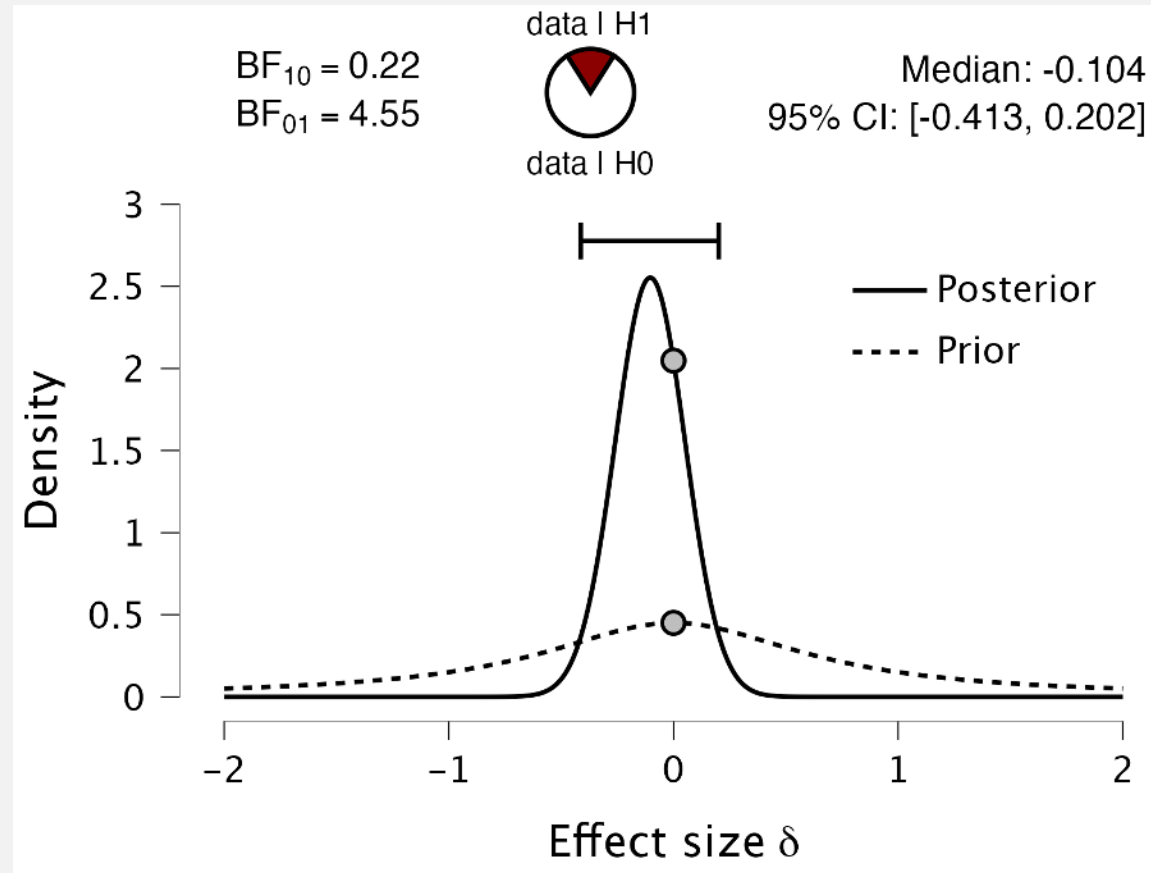


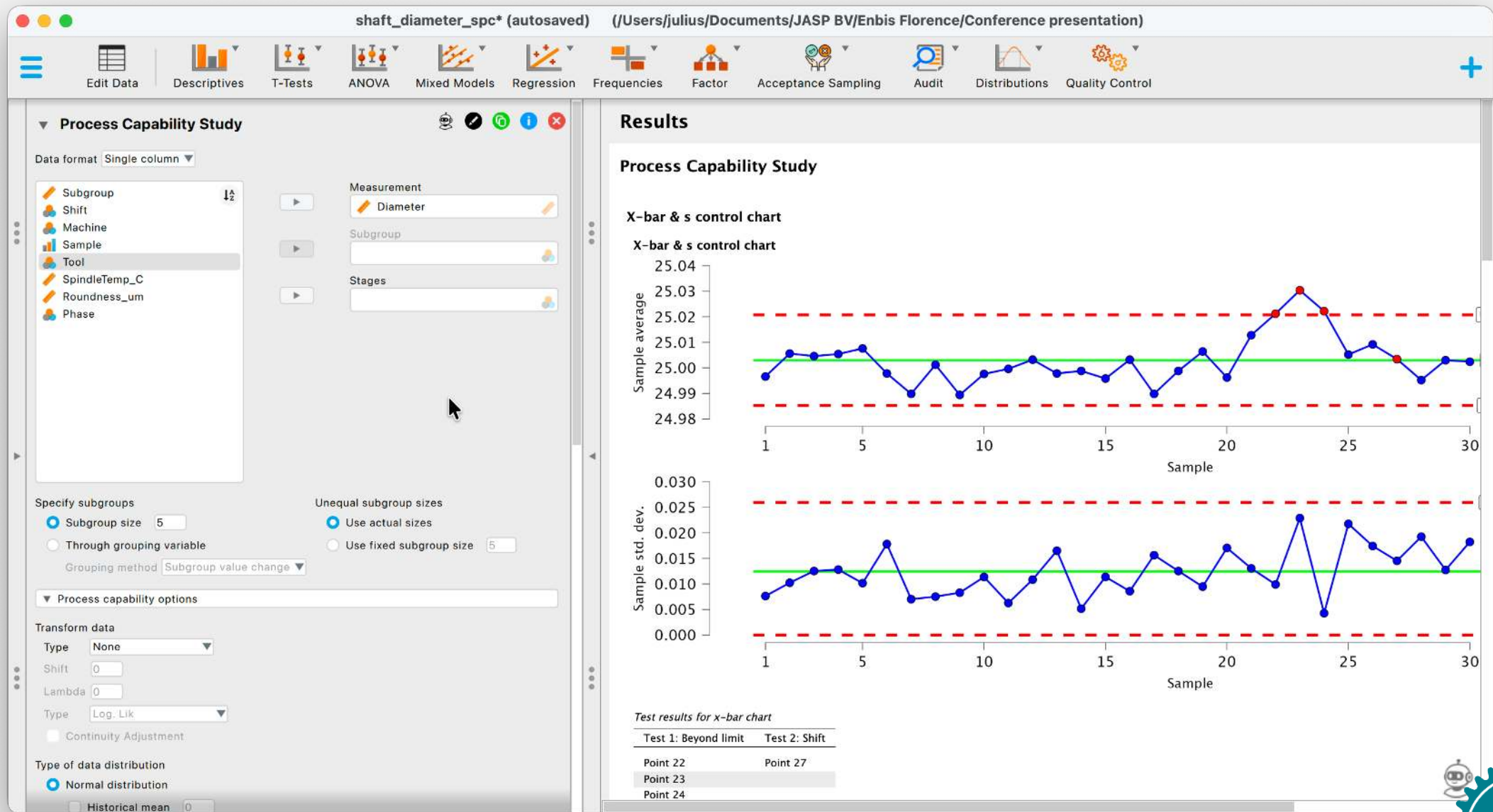
Scatter Plots ▼

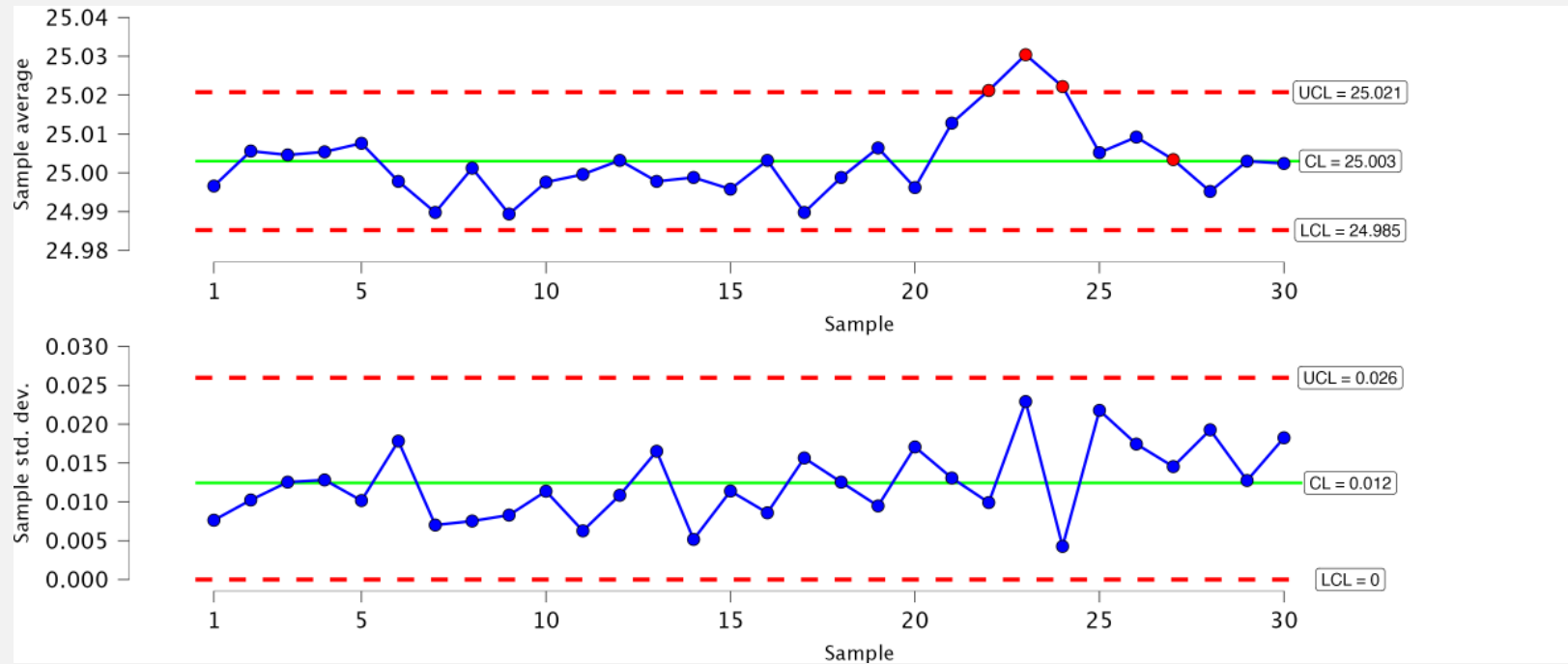


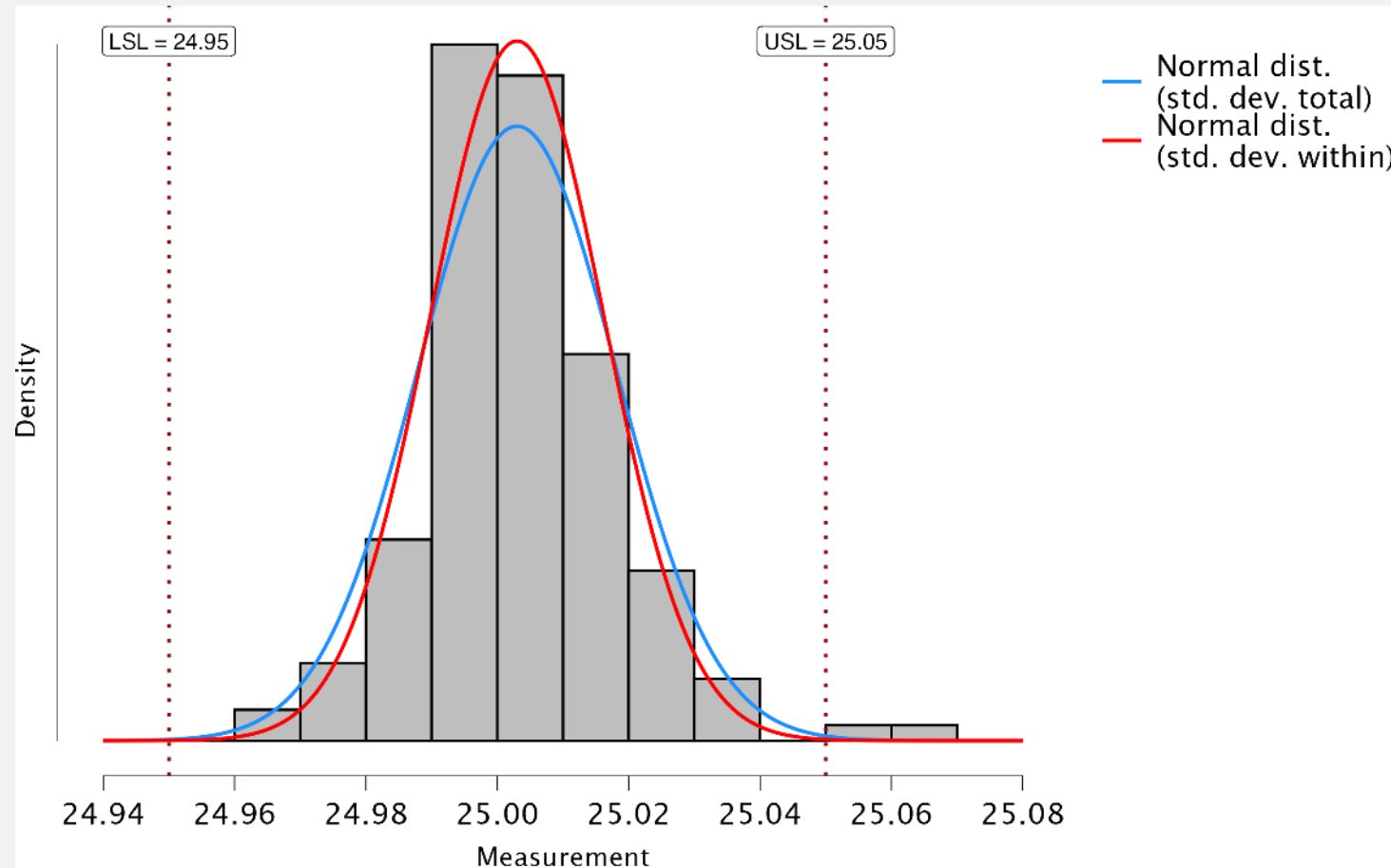


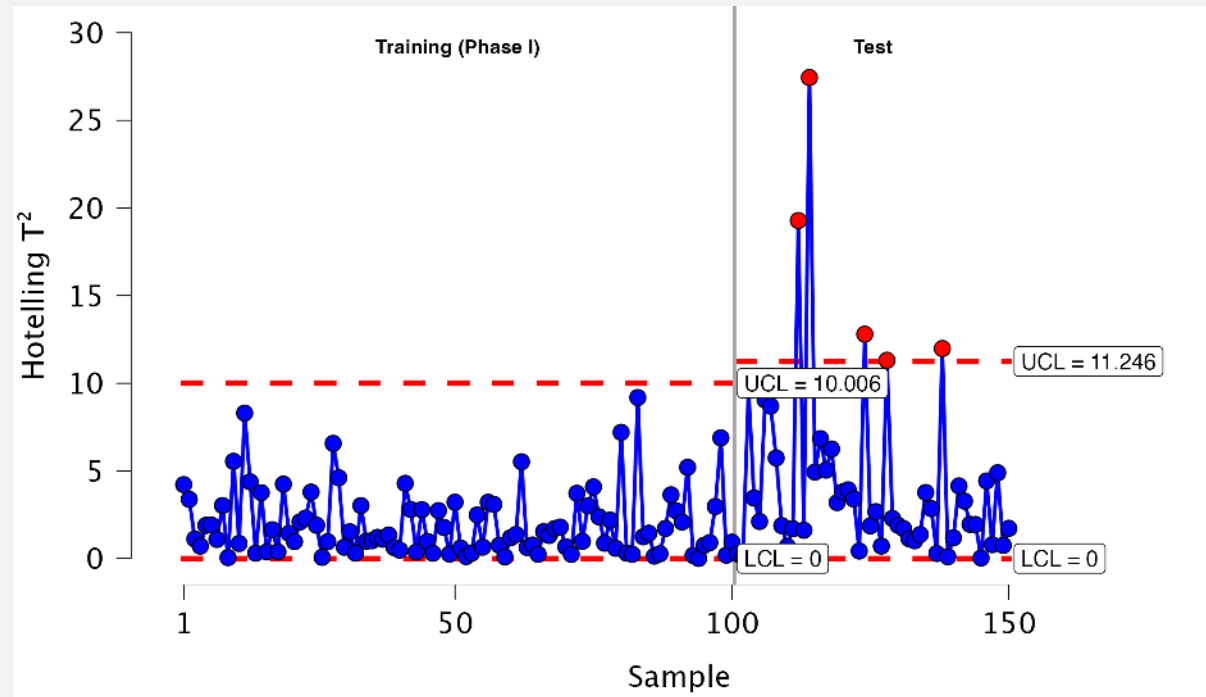






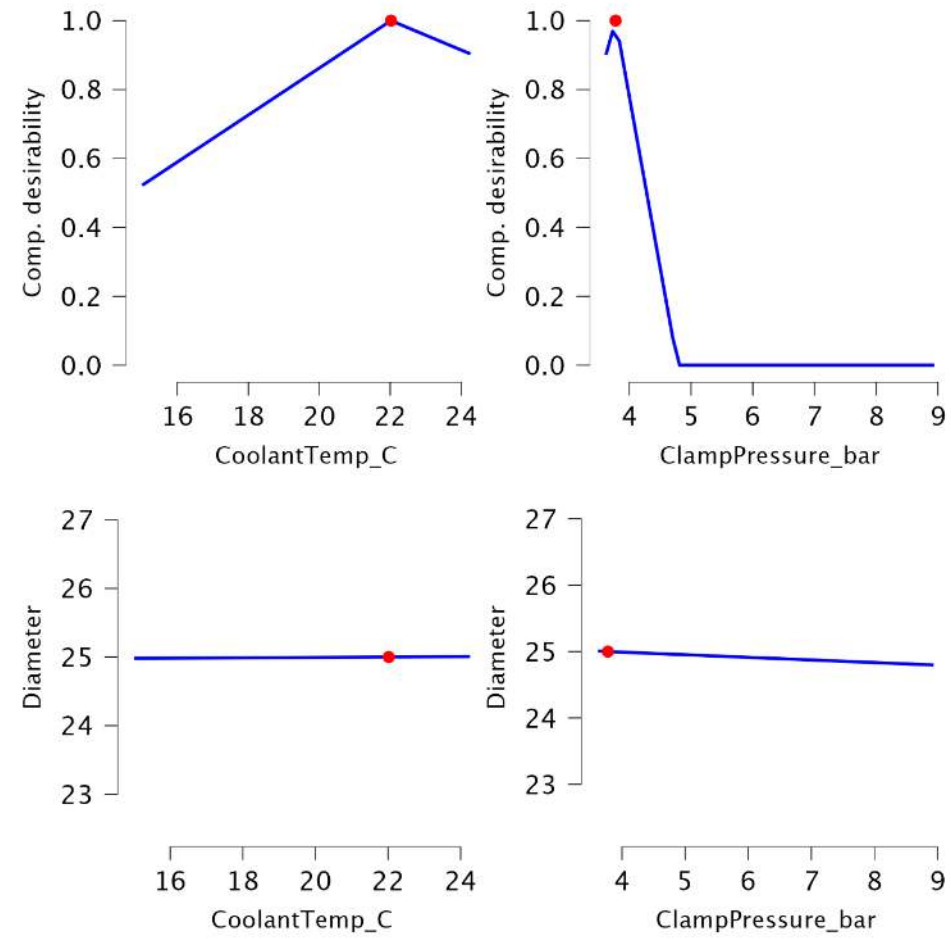






Optimization Plot ▼

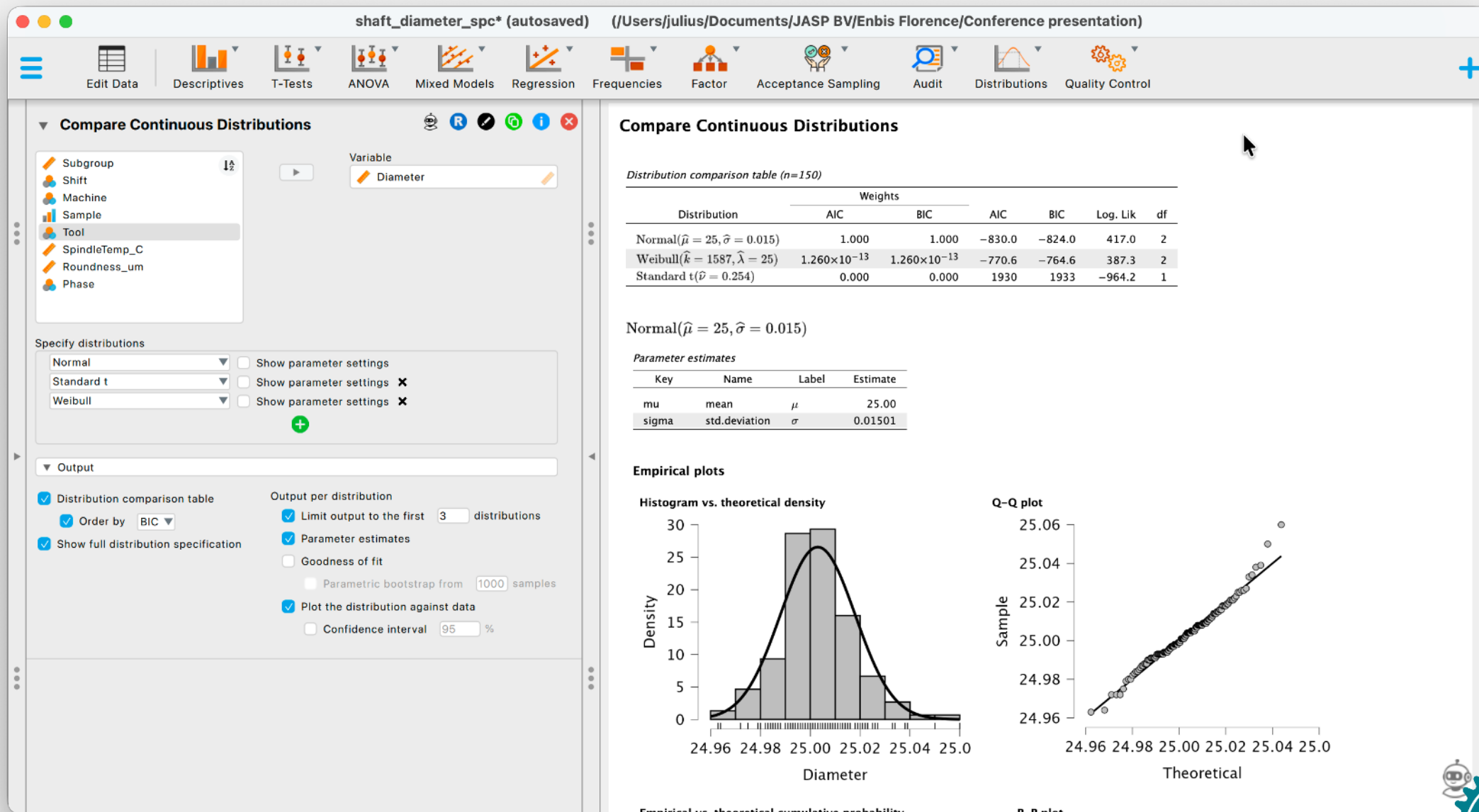
Summary Plot





Compare Distributions in JASP





Normal
Standard normal
Standard t
Student t
Noncentral t
Noncentral Student t
Cauchy
Gumbel
Laplace
Logistic
Skewed generalized t
Symmetric generalized normal
Skew normal
Skew Cauchy
Skew t
Beta
Beta prime
F
Noncentral F
Chi-squared
Noncentral chi-squared
Exponential
Gamma
Inverse gamma
Gompertz
Log-logistic
Log-normal
Wald
Weibull
Amoroso
Stretched beta
Fréchet
Pareto
Triangular
Uniform
Shifted Exponential
Shifted Log-normal
Shifted Gamma
Shifted Inverse gamma
Shifted Log-logistic
Shifted Wald
Shifted Weibull

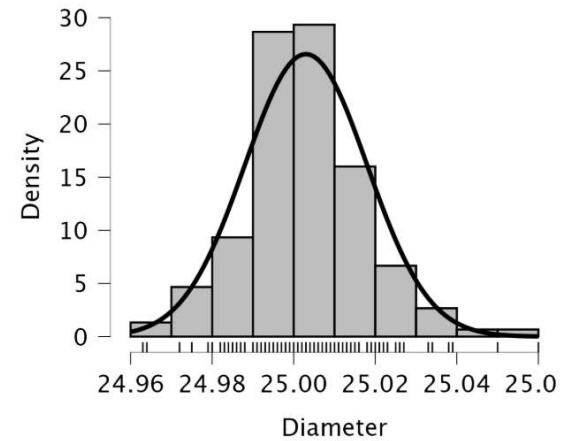


Distribution comparison table (n=150)

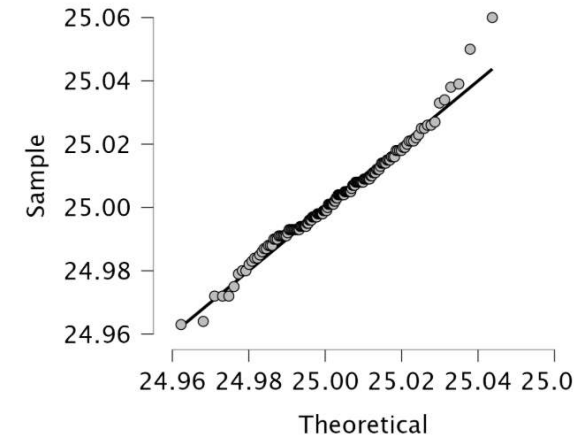
Distribution	Weights		AIC	BIC	Log. Lik	df
	AIC	BIC				
Normal($\hat{\mu} = 25, \hat{\sigma} = 0.015$)	1.000	1.000	-830.0	-824.0	417.0	2
Weibull($\hat{k} = 1587, \hat{\lambda} = 25$)	1.260×10^{-13}	1.260×10^{-13}	-770.6	-764.6	387.3	2
Standard t($\hat{\nu} = 0.254$)	0.000	0.000	1930	1933	-964.2	1



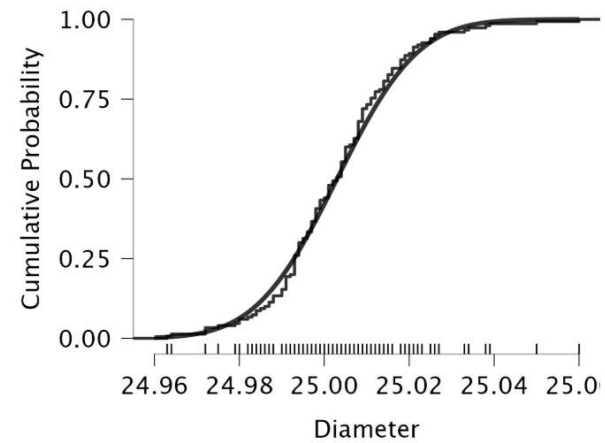
Histogram vs. theoretical density



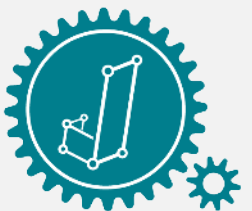
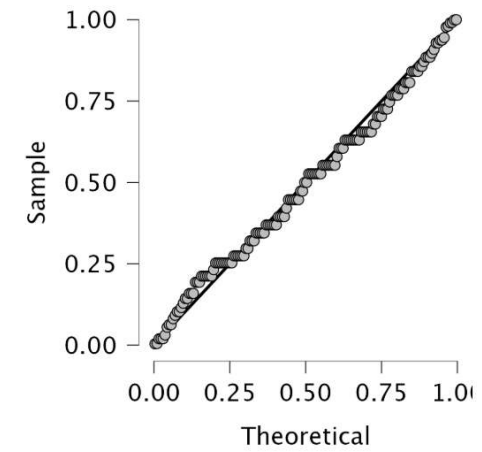
Q-Q plot



Empirical vs. theoretical cumulative probability

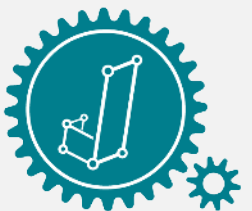


P-P plot





Bayesian Capability Estimation



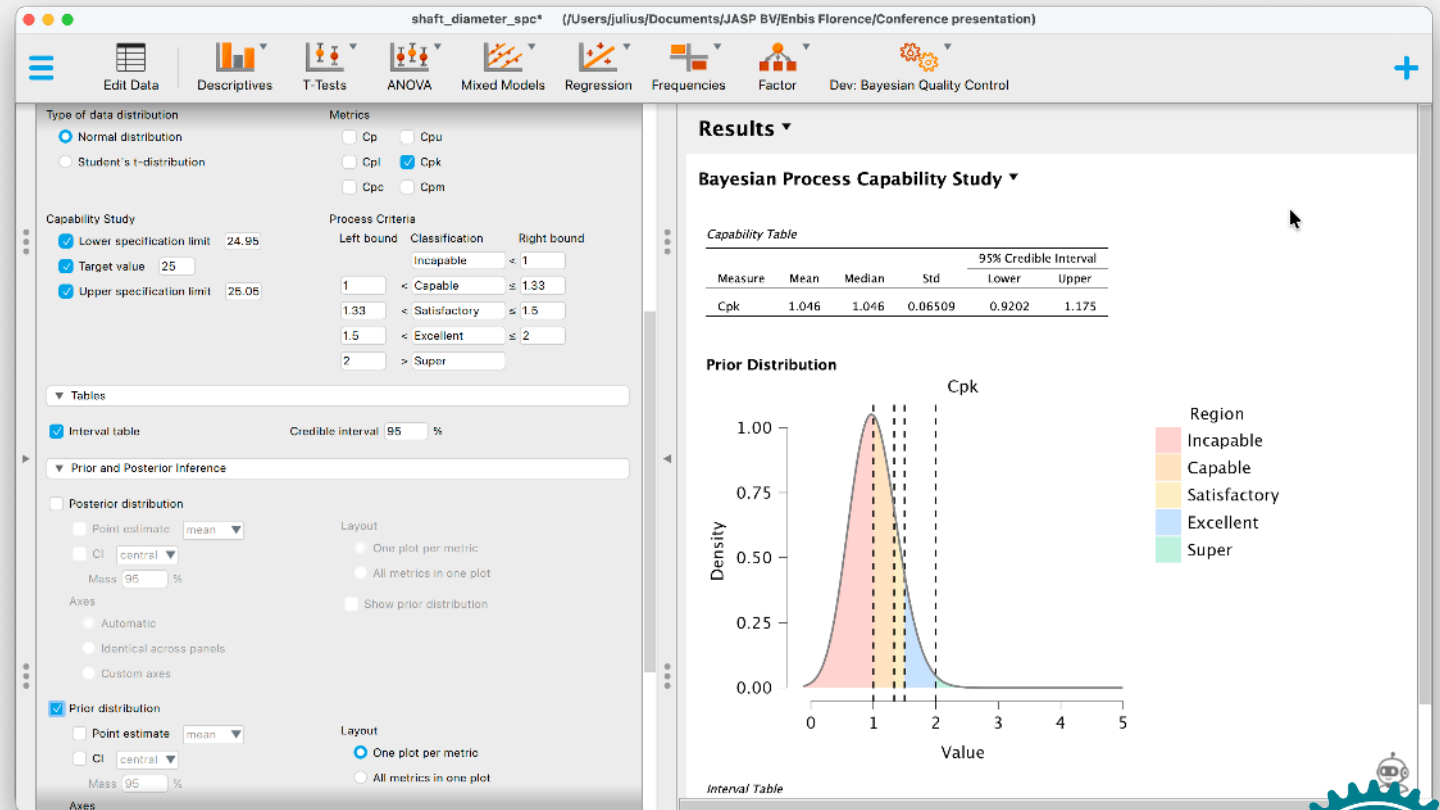
Example: Bayesian Process Capability Analysis

- Shaft production process: Target diameter 25mm
- Is the process within spec?
- Measured by, e.g., C_p , or C_{pk}
- Bayesian inference:
 - Interval that contains C_{pk} with 95% probability (credible interval)
 - Probability that C_{pk} is, e.g, in the “capable” region
 - Probability C_{pk} is larger than, e.g., 1.33
 - Monitor C_{pk} as the data come in



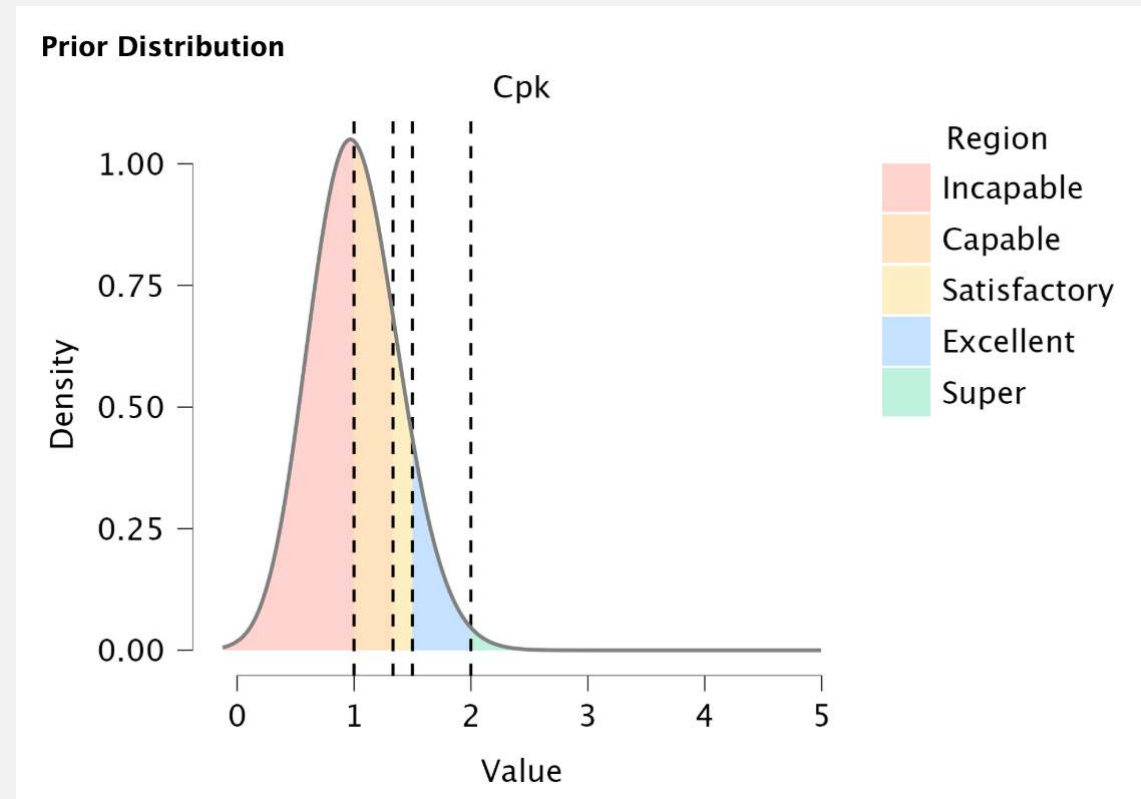
Example: Bayesian Process Capability Analysis

- Assume that the process is normally distributed
- Normal-Inverse- χ^2 prior

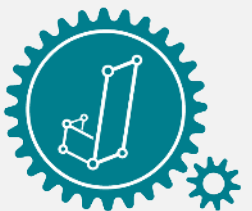
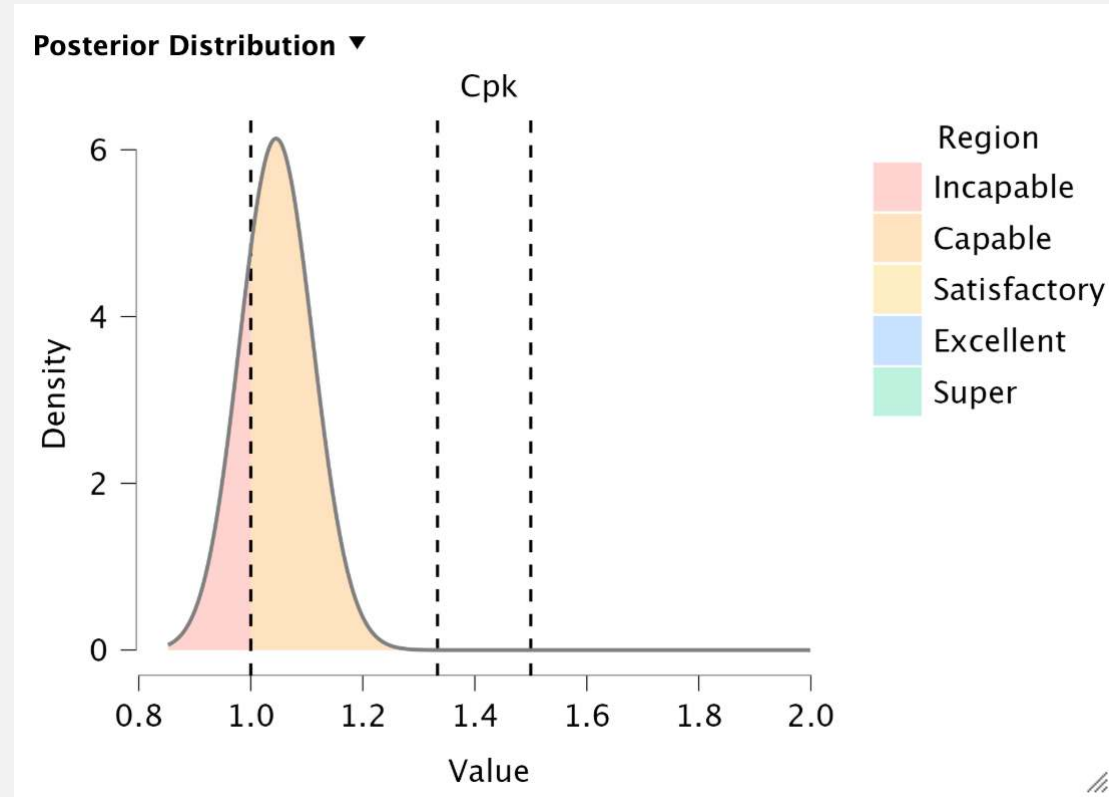


Example: Bayesian Process Capability Analysis

- Assume that the process is normally distributed
- Normal-Inverse- χ^2 prior



Example: Bayesian Process Capability Analysis



Example: Bayesian Process Capability Analysis

Interval Table

Capability Measure	Incapable $(-\text{Inf}, 1]$	Capable $[1, 1.33]$	Satisfactory $[1.33, 1.5]$	Excellent $[1.5, 2]$	Super $[2, \text{Inf})$
Cpk	0.2387	0.7613	1.079×10^{-5}	6.796×10^{-12}	0.000



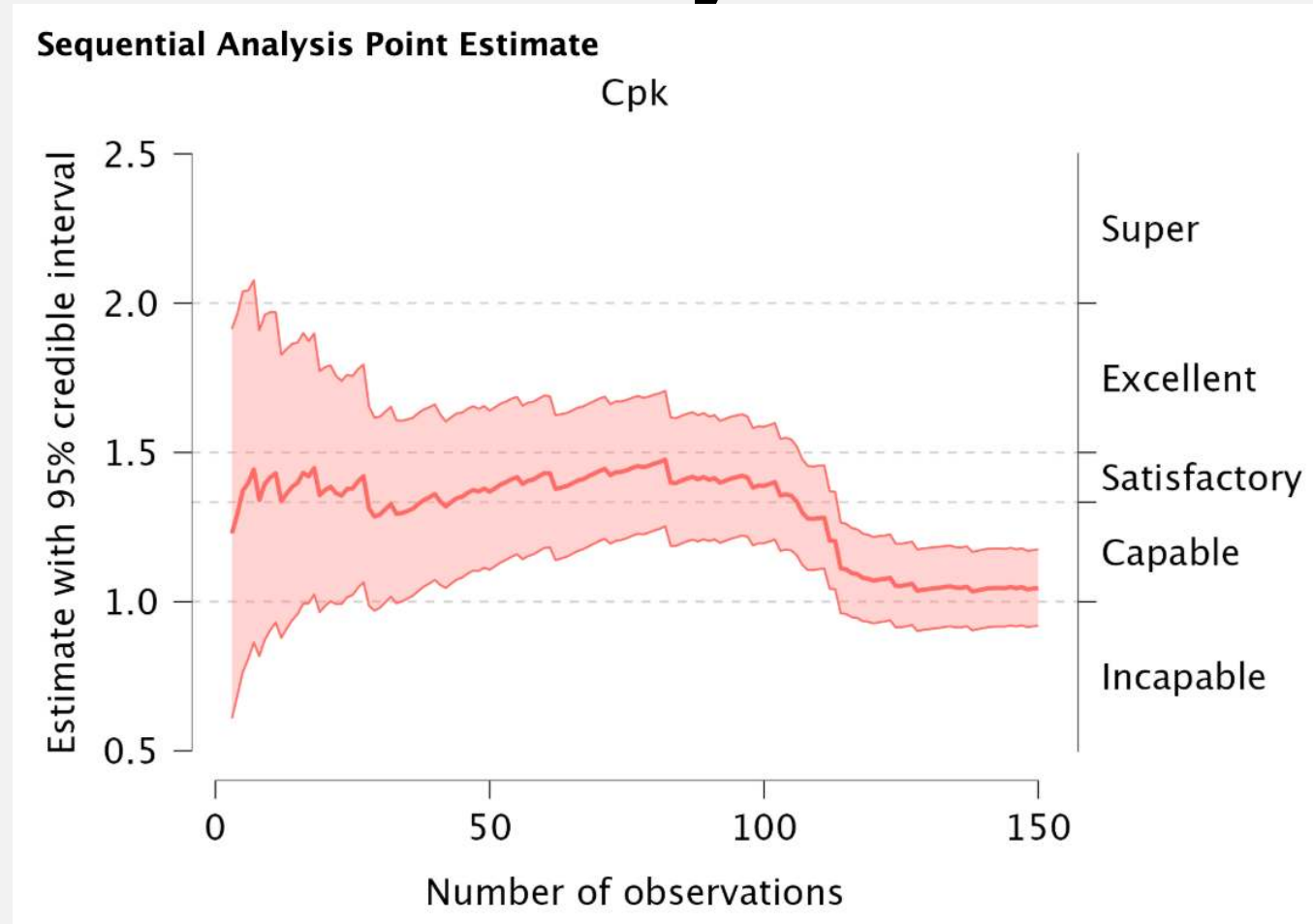
Example: Bayesian Process Capability Analysis

Capability Table ▼

Measure	Mean	Median	Std	95% Credible Interval	
				Lower	Upper
Cpk	1.046	1.046	0.06509	0.9202	1.175



Example: Bayesian Process Capability Analysis





JASP AI

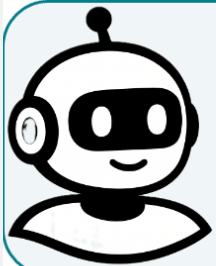
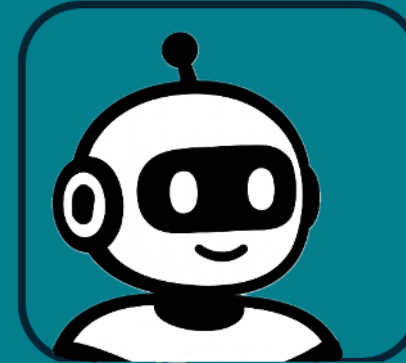


JASP AI

Meet JASP AI

Your statistical assistant inside JASP

Choose, run, explain, teach and report analyses – all inside the JASP workflow



Alfred

The Advisor

Choose, run, interpret, and report analyses.



Socrates

The Tutor

Questions and hints for deeper learning; avoids direct answers.



Evelyn

The Explainer

Direct, step-by-step explanations of concepts and output.



JASP AI

shaft_diameter_spc (/Users/julius/Documents/JASP BV/Enbis Florence/Conference presentation)

Edit Data Descriptives T-Test

JASP AI Chat — Beta Version

Hello, I'm Alfred, your statistical advisor. I'm here to help you with:

- **Choosing** the right statistical analysis for your research question
- **Conducting** analyses in JASP
- **Interpreting** results and checking assumptions
- **Reporting** findings clearly and accurately

To get started, you can describe your research question or data, or ask me anything about statistics or JASP. What can I help you with?

Ask anything...

921 / 921 tokens

Please monitor the costs and always question AI output.
Intended for personal use. Contact [JASP Services BV](#) for enterprise / institutional support.

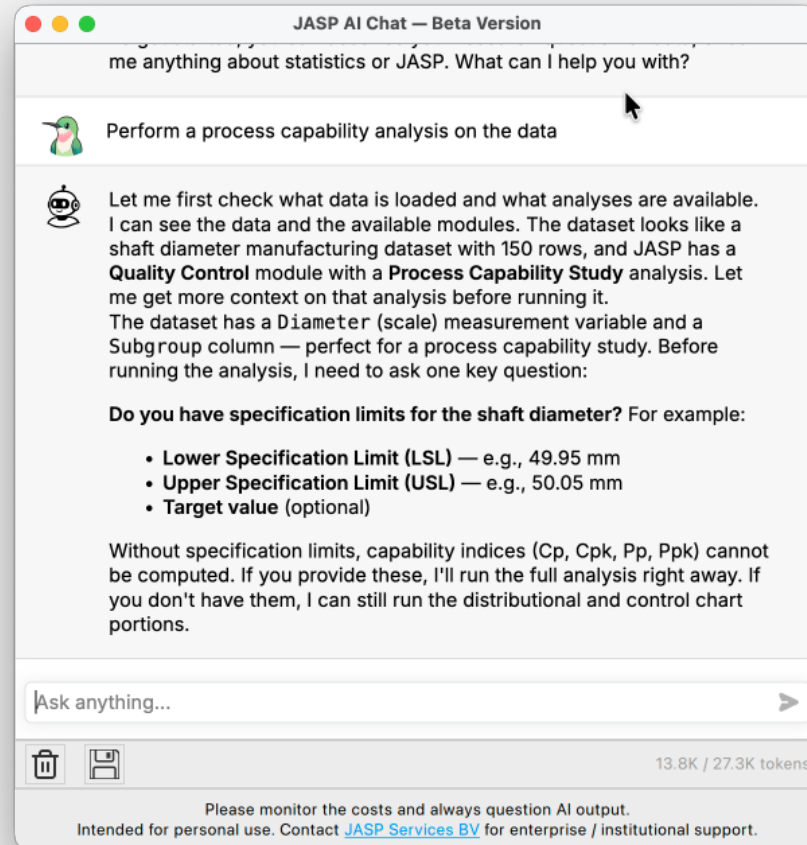
	Subgroup	Shift	Machine	S
1	1	Early	M-01	1
2	1	Early	M-01	2
3	1	Early	M-01	3
4	1	Early	M-01	4
5	1	Early	M-01	5
6	2	Late	M-02	1
7	2	Late	M-02	2
8	2	Late	M-02	3
9	2	Late	M-02	4
10	2	Late	M-02	5
11	3	Night	M-01	1
12	3	Night	M-01	2
13	3	Night	M-01	3
14	3	Night	M-01	4
15	3	Night	M-01	5
16	4	Early	M-02	1
17	4	Early	M-02	2
18	4	Early	M-02	3
19	4	Early	M-02	4
20	4	Early	M-02	5
21	5	Late	M-01	1

25.008 Insert-A 23.81 3.47 Phase I

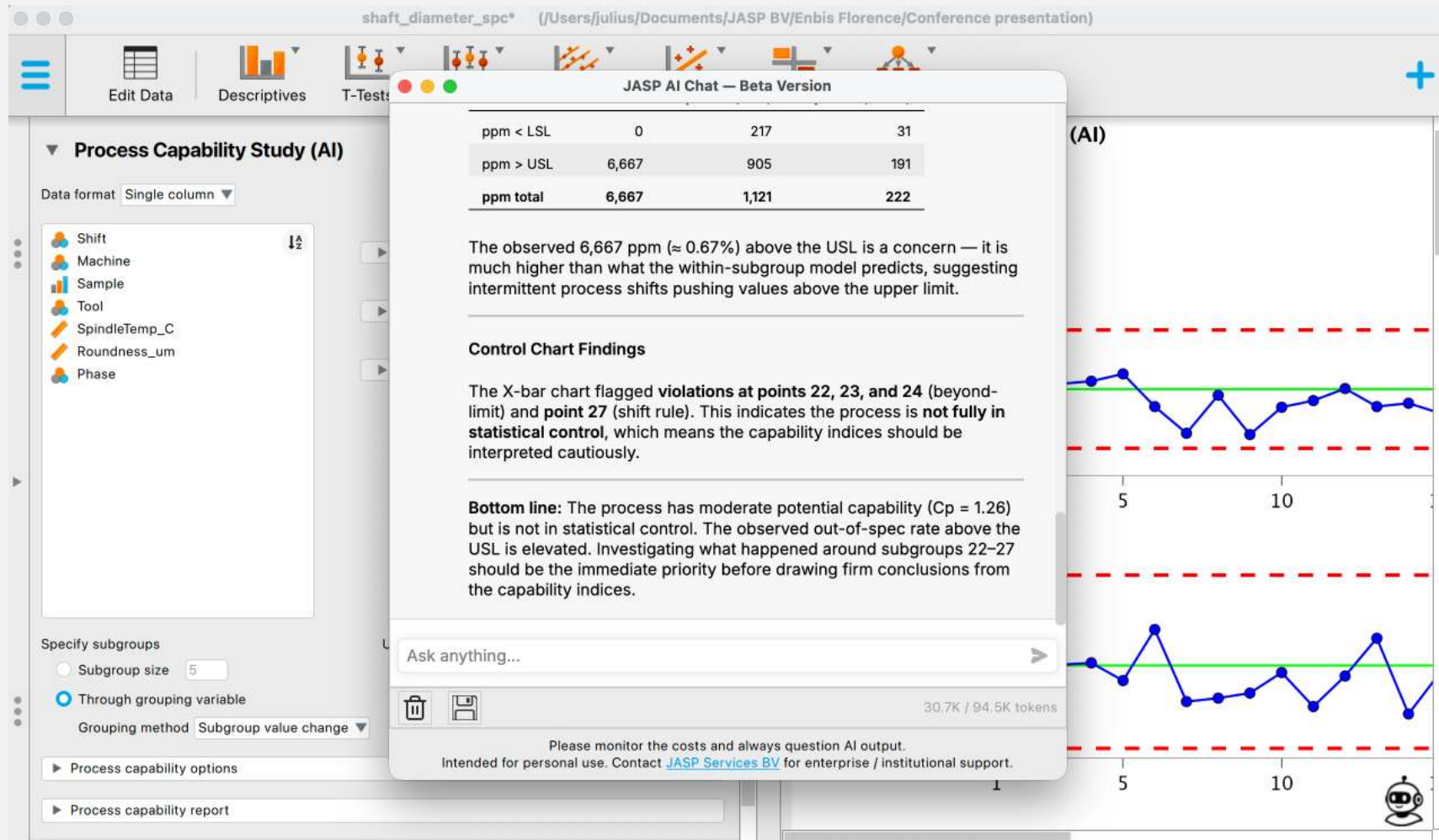
24.001 Insert-A 21.00 3.07 Phase I



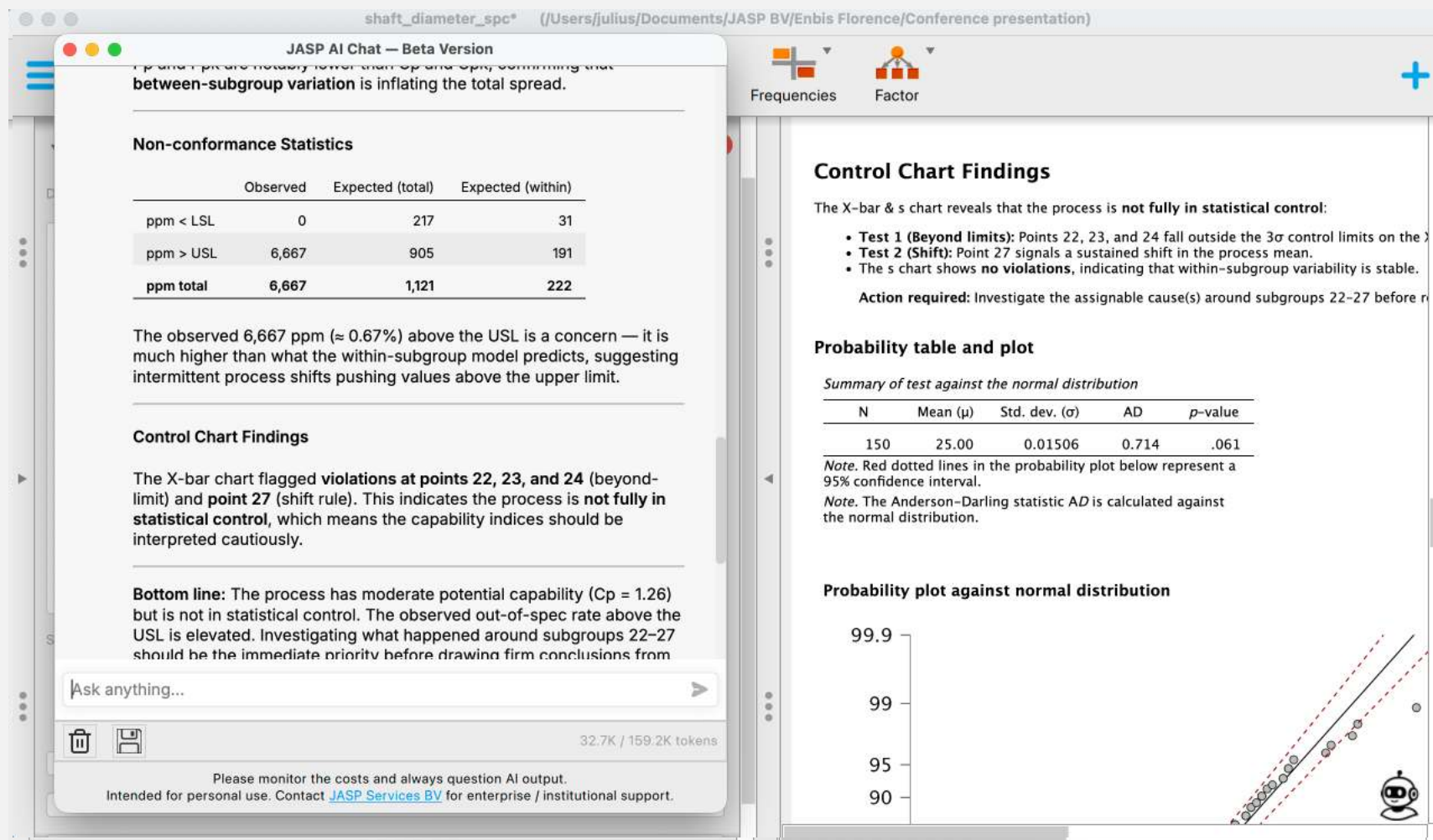
JASP AI



JASP AI



JASP AI



Conclusion

- JASP is free and open source
- Feedback is welcome



<https://jasp-stats.org>



<https://www.jasp-services.com>

Thank you!